

JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles.

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JASPAR (<https://jaspar.elixir.no/>) is a widely-used open-access database presenting manually curated high-quality and non-redundant DNA-binding profiles for transcription factors (TFs) across taxa. In this 10th release and 20th-anniversary update, the CORE collection has expanded with 329 new profiles. We updated three existing profiles and provided orthogonal support for 72 profiles from the previous release's UNVALIDATED collection. Altogether, the JASPAR 2024 update provides a 20% increase in CORE profiles from the previous release. A trimming algorithm enhanced profiles by removing low information content flanking base pairs, which were likely uninformative (within the capacity of the PFM models) for TFBS predictions and modelling TF-DNA interactions. This release includes enhanced metadata, featuring a refined classification for plant TFs' structural DNA-binding domains. The new JASPAR collections prompt updates to the genomic tracks of predicted TF binding sites (TFBSs) in 8 organisms, with human and mouse tracks available as native tracks in the UCSC Genome browser. All data are available through the JASPAR web interface and programmatically through its API and the updated Bioconductor and pyJASPAR packages. Finally, a new TFBS extraction tool enables users to retrieve predicted JASPAR TFBSs intersecting their genomic regions of interest.

Oral Antioxidant and Lutein/Zeaxanthin Supplements Slow Geographic Atrophy Progression to the Fovea in Age-Related Macular Degeneration.

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To determine whether oral micronutrient supplementation slows geographic atrophy (GA) progression in age-related macular degeneration (AMD). Post hoc analysis of Age-Related Eye Disease Study (AREDS) and AREDS2, multicenter randomized placebo-controlled trials of oral micronutrient supplementation, each with 2 × 2 factorial design. A total of 392 eyes (318 participants) with GA in AREDS and 1210 eyes (891 participants) with GA in AREDS2. The AREDS participants were randomly assigned to oral antioxidants (500 mg vitamin C, 400 IU vitamin E, 15 mg β-carotene), 80 mg zinc, combination, or placebo. The AREDS2 participants were randomly assigned to 10 mg lutein/2 mg zeaxanthin, 350 mg docosahexaenoic acid/650 mg eicosapentaenoic acid, combination, or placebo. Consenting AREDS2 participants were also randomly assigned to alternative AREDS formulations: original; no beta-carotene; 25 mg zinc instead of 80 mg; both. (1) Change in GA proximity to central macula over time and (2) change in square root GA area over time, each measured from color fundus photographs at annual visits and analyzed by mixed-model regression according to randomized assignments. In AREDS eyes with noncentral GA (n = 208), proximity-based progression toward the central macula was significantly slower with randomization to antioxidants versus none, at 50.7 μm/year (95% confidence interval [CI], 38.0-63.4 μm/year) versus 72.9 μm/year (95% CI, 61.3-84.5 μm/year; P = 0.012), respectively. In AREDS2 eyes with noncentral GA, in participants assigned to AREDS antioxidants without β-carotene (n = 325 eyes), proximity-based progression was significantly slower with randomization to lutein/zeaxanthin versus none, at 80.1 μm/year (95% CI, 60.9-99.3 μm/year) versus 114.4 μm/year (95% CI, 96.2-132.7 μm/year; P = 0.011), respectively. In AREDS eyes with any GA (n = 392), area-based progression was not significantly different with randomization to antioxidants versus none (P = 0.63). In AREDS2 eyes with any GA, in participants assigned to AREDS antioxidants without β-carotene (n = 505 eyes), area-based progression was not significantly different with randomization to lutein/zeaxanthin versus none (P =

0.64). Oral micronutrient supplementation slowed GA progression toward the central macula, likely by augmenting the natural phenomenon of foveal sparing. Proprietary or commercial disclosure may be found after the references.

Ophthalmologists' awareness of geographic atrophy: An Italian survey including 365 participants.

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Geographic atrophy (GA) is a severe complication of age-related macular degeneration (AMD) and leads to irreversible visual decline. To date, no effective treatment is available for GA patients. However, a number of new therapies have recently been approved and several others are in the pipeline. This rapid evolution of prospects for GA patients requires constant updating of ophthalmologists' understanding of GA and its management so as to provide the appropriate treatment. For this reason, Società Italiana di Scienze Oftalmologiche (S.I.S.O.) has designed a specific survey to gauge the position of Italian ophthalmologists in this regard. The three hundred and sixty-five Italian ophthalmologists who agreed to take part received a seventeen-part questionnaire guaranteeing privacy and anonymity. The survey was compiled through an online portal and the results were sent directly to S.I.S.O. ETS. Two graders analyzed the data and recorded the results. The results showed a high level of self-assessed awareness and understanding of GA, as well as considerable willingness to further improve knowledge of the disease. Most of the participants claimed to have effective rules of conduct in place for managing GA patients, including prompt response, involving a high prevalence of nutraceutical prescriptions and lifestyle recommendations. This survey provided an overview of how GA patients are managed in Italy. The Italian ophthalmology community appears to be ready to adopt the upcoming treatments for GA.

The Discrepancy Between Visual Acuity Decline and Foveal Involvement in Geographic Atrophy.

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To investigate the discrepancy between visual acuity (VA) decline and foveal involvement in geographic atrophy (GA) secondary to nonexudative age-related macular degeneration (AMD), and to explore how early retinal changes impact the progression of visual impairment. Retrospective, longitudinal cohort study. This study evaluated 80 eyes from 60 patients (mean age, 74.2 ± 10 years) with progressing non-neovascular AMD. Blue-light fundus autofluorescence (FAF) and spectral-domain OCT (SD-OCT) were utilized to monitor GA progression and the onset of foveal involvement. The study analyzed VA changes over an average follow-up of 60 ± 26.4 months, encompassing 785 observations. Mixed-effects models with natural splines assessed the effects of demographic and ocular characteristics on baseline VA and its rate of decline. Survival analyses compared the timing of anatomic changes with the most rapid functional declines, indicated by the highest first derivative of VA trajectories. Discrepancies between visual and anatomic changes were explored using generalized linear mixed-effects models. Monthly VA changes, onset and impact of foveal involvement, and factors influencing baseline VA and rate of decline. Visual acuity declined consistently by an average of 0.010 logarithm of the minimum angle of resolution (LogMAR) per month (standard error [SE], 0.0003; $P < 0.001$). The onset of foveal involvement significantly exacerbated this decline, adding an average loss of 0.15 LogMAR (SE, 0.02; $P < 0.001$).

Stabilization of VA typically occurred around 41 months post-foveal involvement. Significant factors associated with worse baseline VA were older age, female gender, unifocal GA morphology, and drusen-associated forms of GA ($P < 0.05$). The most rapid declines in VA typically occurred about 9 months (interquartile range, 0-27 months) before detectable subfoveal changes. The reticular FAF pattern (27/46 [59%] vs. 2/13 [15%], $P = 0.02$) and smaller baseline GA lesions ($P = 0.01$) were associated with faster deterioration preceding visible foveal damage. This study demonstrates that significant VA loss in GA can precede detectable foveal involvement, suggesting a window for early interventions to slow the progression of visual impairment. Identifying specific GA characteristics and FAF patterns as predictors of rapid VA decline supports the need for personalized treatment strategies to optimize outcomes for patients with nonexudative AMD. Proprietary or commercial disclosure may be found in the Footnotes and Disclosures at the end of this article.

Geographic atrophy and factors associated with disease progression among Medicare Advantage enrollees.

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Geographic atrophy (GA) is a form of advanced age-related macular degeneration (AMD) that can cause irreversible vision impairment and is responsible for approximately 20% of legal blindness in the United States. There is limited real-world evidence assessing health outcomes and health care resource use (HCRU) among individuals with GA. To examine the progression from GA without subfoveal involvement (SFI) to GA with SFI, progression to irreversible blindness, and HCRU among older individuals with GA enrolled in Medicare Advantage Prescription Drug (MAPD) plans. This retrospective study used claims data for MAPD-plan enrollees aged at least 65 years with an AMD diagnosis between 2018 and 2021. To assess progression of GA, development of blindness, and HCRU, propensity score matched cohorts of individuals with GA and without GA were identified and compared. For GA progression analysis, at least 12 months of follow-up was required, and patients were followed until the end of either follow-up or study period. Total 9,511 individuals with GA were matched 1:1 to individuals without GA. Among individuals with GA, initial diagnosis was primarily by an ophthalmologist (58.6%) followed by an optometrist (30.9%). The most common diagnostic imaging procedure at index was optical coherence tomography (53.0%). Mean follow-up time was 2.3 years. At index, 4,781 (50.3%) individuals had GA without SFI and 4,697 (49.4%) had GA with SFI. Among individuals with GA without SFI at index, 479 (10.2%) progressed to GA with SFI during the 12-month follow-up. Among individuals with GA without SFI at index, 173 (3.6%) developed irreversible blindness, compared to 312 (6.6%) of those with SFI, and 51 (0.5%) individuals without GA. Kaplan-Meier analysis indicated fastest progression to irreversible blindness among individuals with GA with SFI, followed by those without SFI (log-rank test $P < 0.001$). Both diagnosis of GA without SFI (hazard ratio [HR] [CI] = 6.77 [4.98-9.35], $P < 0.001$) and diagnosis of GA with SFI (HR [CI] = 12.59 [9.43-17.16], $P < 0.001$) were strongly associated with increased risk of developing irreversible blindness. Significant predictors of progression to GA with SFI were wet AMD at baseline (HR [CI] = 5.70 [4.63-6.99], $P < 0.001$), Elixhauser comorbidity score of 4-5 (HR [CI] = 1.46 [1.12-1.91], $P = 0.006$), and more than 5 (HR [CI] = 1.40 [1.02-1.89], $P = 0.035$). GA with or without SFI was associated with progression to irreversible blindness in an MAPD-plan population. Patients with GA with SFI progressed to irreversible blindness faster than patients with GA without SFI. With the recent approval of GA treatments, future research is needed to assess the impacts on disease progression, including blindness.

Incidence and progression of geographic atrophy: observations from a population-based cohort.

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To examine early age-related macular degeneration (AMD) lesion characteristics and risk factors associated with the long-term development and progression of geographic atrophy (GA). Population-based cohort. Of 3654 participants aged ≥ 49 years in the Blue Mountains Eye Study, 75.8%, 76.7%, and 56.1% of survivors attended the 5-, 10-, and 15-year follow-up examinations, respectively. Retinal photographs were taken at each visit. Incident GA was confirmed using a side-by-side grading method. Computer planimetry was used to measure the area involved by GA. Fast and slow/normal progression rates were defined as GA area enlargement by ≥ 2 and < 2 mm²/year, respectively. Incident GA was estimated using the Kaplan-Meier product-limit method. Early AMD lesion characteristics were assessed for association with GA incidence using eye-specific data and generalized estimating equation models adjusting for age, current smoking, and presence of risk alleles of the complement factor H (CFH) or age-related maculopathy susceptibility 2 (ARMS2) genes, genotyped or imputed using genome-wide scan data. Incidence and progression of GA. By excluding 41 subjects with GA at baseline, of 2503 participants at risk of GA, incident pure GA (without coexisting neovascular AMD lesions) was confirmed in 57 participants, with a 15-year incidence of 3.6%. Baseline early AMD lesion characteristics associated with GA incidence included drusen type (soft indistinct: odds ratio [OR], 59.0; 95% confidence interval [CI], 20.4-171.0; reticular drusen: OR, 13.9; 95% CI, 4.0-47.6); drusen location within a 500- μ m radius of the fovea (OR, 15.1; 95% CI, 7.4-30.8); drusen area greater than 375 μ m in diameter (OR, 10.1; 95% CI, 4.0-25.6); presence of retinal pigment epithelial depigmentation (OR, 9.0; 95% CI, 4.1-19.8); or hyperpigmentation (OR, 12.0; 95% CI, 6.1-23.5), referenced to subjects with no or hard drusen only. Fast progression was more frequent among current smokers at baseline, subjects with the CFH or ARMS2 risk genotypes, and pseudophakic eyes. Early AMD lesion characteristics (type, location, area involved) were strongly associated with higher long-term risk of developing GA independent of age, smoking, and AMD genetic susceptibility from the CFH or ARMS2 genes. Known AMD risk factors also were more frequently present among quickly progressing GA cases. Proprietary or commercial disclosure may be found after the references.

Designing clinical trials for age-related geographic atrophy of the macula: enrollment data from the geographic atrophy natural history study.

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To derive information from the Geographic Atrophy (GA) Natural History Study that is relevant to recruiting patients and designing clinical trials for GA. A prospective natural history study with annual follow-up enrolled patients with GA and no choroidal neovascularization (CNV) in at least one eye. Characteristics of recruited and enrolled patients are analyzed, in the context of progression data from the study. The data show that GA from age-related macular degeneration (AMD) was seen in 82% of the referred patients, there was an attrition rate of 14%, and 60% of the patients with GA from AMD had bilateral GA without CNV. Within the 83 patients in the bilateral GA group with follow-up, 50 patients (60%) met both the proposed visual acuity and the proposed GA area criteria for a treatment trial in one or both eyes. These data should be helpful in planning future treatment trials for GA.

Geographic Atrophy and OCT Angiography: Descriptive Study and Correlation With Autofluorescence.

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Geographic atrophy (GA) involves the progressive loss of retinal pigment epithelium (RPE), photoreceptors, and choriocapillaris (CC). CC flow within a GA area is severely impaired in patients with atrophic age-related macular degeneration. The aim of this study was to compare GA area measured on optical coherence tomography angiography (OCTA) (CC nonperfusion area) and on fundus autofluorescence (FAF). In this prospective, observational, cross-sectional study, OCTA and FAF were performed in patients with GA. On OCTA (CC segmentation), the CC nonperfusion area was measured manually using calipers. On FAF, GA was manually delimited, and the total surface was obtained using Region Finder software. The primary endpoint was to compare the CC nonperfusion area measured on OCTA and on the gold standard method (FAF). Forty eyes of 34 patients with a mean age of 82.63 years $\pm$ 9.21 years (range: 66 years to 100 years) were included. The mean GA area measured on FAF and OCTA was, respectively, 2.184 $\pm$ 3.045 mm² and 2.349 $\pm$ 3.237 mm² ($P = .035$). The mean difference was 0.165 $\pm$ 0.290 mm². A strong correlation was found between both measurements ($r = 0.97$; $P < .0001$; confidence interval: 0.98-0.99), although the CC nonperfusion area was larger than the GA area on FAF ($P = .035$). In this study, the authors showed that in GA, the CC nonperfusion area correlates linearly with the GA area assessed by FAF. Also, the CC nonperfusion area is larger than the GA area measured by FAF, suggesting that CC degeneration could occur before RPE degeneration in GA. [Ophthalmic Surg Lasers Imaging Retina. 2019;50:e222-e228.].

GEOGRAPHIC ATROPHY: Semantic Considerations and Literature Review.

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There is a lack of agreement regarding the types of lesions and clinical conditions that should be included in the term "geographic atrophy." Varied and conflicting views prevail throughout the literature and are currently used by retinal experts and other health care professionals. We reviewed the nominal definition of the term "geographic atrophy" and conducted a search of the ophthalmologic literature focusing on preceding terminologies and the first citations of the term "geographic atrophy" secondary to age-related macular degeneration. According to the nominal definition, the term "geography" stands for a detailed description of the surface features of a specific region, indicating its relative position. However, it does not necessarily imply that the borders of the region must be sharply demarcated or related to any anatomical structures. The term "geographical areas of atrophy" was initially cited in the 1960s in the ophthalmologic literature in the context of uveitic eye disease and shortly thereafter also for the description of variants of "senile macular degeneration." However, no direct explanation could be found in the literature as to why the terms "geographical" and "geographic" were chosen. Presumably the terms were used as the atrophic regions resembled the map of a continent or well-defined country borders on thematic geographical maps. With the evolution of the terminology, the commonly used adjunct "of the retinal pigment epithelium" was frequently omitted and solely the term "geographic atrophy" prevailed for the nonexudative late-stage of age-related macular degeneration itself. Along with the quantification of atrophic areas, based on different imaging modalities and the use of both manual and semiautomated approaches, various and inconsistent definitions for the minimal lesion diameter or size of atrophic lesions have also emerged. Reconsideration of the application of the term "geographic atrophy" in the context of age-related macular degeneration seems to be prudent given ongoing advances in multimodal retinal imaging technology with identification of various phenotypic

characteristics, and the observation of atrophy development in eyes under antiangiogenic therapy.

Geographic atrophy: a histopathological assessment.

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Geographic atrophy (GA) is the major cause of blind registration in Western communities, although, with few exceptions, it is less common than choroidal neovascular disease. The variation of phenotype implies that age-related macular degeneration (AMD) does not follow the same course from one case to another and that phenotyping may be important before initiating a therapeutic trial. To document photoreceptor and retinal pigment epithelium (RPE) cell loss and other changes at the RPE-choroid interface in donated human eyes in which visual loss was deemed to be due to GA. Histological study of a consecutive series of eyes donated by individuals previously diagnosed clinically as having GA. Donors were chosen on the basis of available clinical records (from MidAmerica Transplant Services, St Louis, Missouri; the Iowa Lions Eye Bank, Iowa City; and the Utah Lions Eye Bank, Salt Lake City) and selected were those considered to have GA due to AMD. Tissues in the regions of atrophy were examined with light, electron, and autofluorescence microscopy. In most of the 37 donors examined, there was marked loss of photoreceptor cells for variable distances distal from the edge of the GA. Rod loss was greater than cone loss. An inverse relationship existed between the quantity of autofluorescent inclusions in the RPE and the thickness of sub-RPE basal laminar deposit. Integrity of the choroid varied from one eye to another and was not related strictly to photoreceptor survival. In some eyes, photoreceptor loss existed in the absence of obvious morphological changes in the Bruch membrane or RPE. The findings support the view that photoreceptor loss occurs early in AMD in a proportion of cases and imply that photoreceptor-cell loss may contribute to the functional loss recorded in early stages of AMD at least in part. The variation of changes from one eye to another implies that patients selected for a specific prophylactic therapy for early AMD should be chosen on the basis of the characteristics of their disease.
